UNIQUENESS OF THE AMELIORATION OF A 1-MORPHISM OF CATEGORIES FIBRED IN GROUPOIDS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7793-7844, following the statement at 7771-7791.

Editorial note. The identification of the comparison object uses the in-chapter construction referenced explicitly; all 2-isomorphism data is displayed in-span.

Allowed standard prerequisites (editorial).

- 2-fibre product of categories over C. Lemma lemma-2-product-categories-over-C of this chapter; the identification of the ameliorated category with X x_{F,Y,id} Y is given in the proof of lemma-ameliorate-morphism-categories-fibred-groupoids at 7725-7728. The comparison object over Y to which both X' and X" are related.
- Equivalences and 2-fibre products. Lemma lemma-equivalence-2-fibre-product of this chapter, invoked at proof lines 7813-7814. c is an equivalence because b is.
- Equivalences of categories fibred in groupoids. Lemma lemma-equivalence-fibred-categories of this chapter, invoked at proof lines 7815-7817. Promotes the equivalence over C to an equivalence in the 2-category of categories fibred in groupoids over Y.

Source reference. Original lemma 06N7: sources/stacks-categories/pinned-original-source.tex, source lines 7707–7707. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02XB: sources/stacks-categories/pinned-original-source.tex, source lines 5933–5933. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 003Z: sources/stacks-categories/pinned-original-source.tex, source lines 7488–7488. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $F : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of categories fibred in groupoids over $\mathcal{C}$. Assume we have a 2-commutative diagram*

$$\begin{array}{ccccc} \mathcal{X}' & \xleftarrow{a} & \mathcal{X} & \xrightarrow{b} & \mathcal{X}'' \\ & {\scriptstyle f}\searrow & \downarrow{\scriptstyle F} & \swarrow{\scriptstyle g} & \\ & & \mathcal{Y} & & \end{array}$$

where a and b are equivalences of categories over $\mathcal{C}$ and f and g are categories fibred in groupoids. Then there exists an equivalence $h : \mathcal{X}'' \to \mathcal{X}'$ of categories over $\mathcal{Y}$ such that $h \circ b$ is 2-isomorphic to a as 1-morphisms of categories over $\mathcal{C}$. If the diagram above actually commutes, then we can arrange it so that $h \circ b$ is 2-isomorphic to a as 1-morphisms of categories over $\mathcal{Y}$.

Proof. We will show that both $\mathcal{X}'$ and $\mathcal{X}''$ over $\mathcal{Y}$ are equivalent to the category fibred in groupoids $\mathcal{X} \times_{F,\mathcal{Y},\mathrm{id}} \mathcal{Y}$ over $\mathcal{Y}$, see proof of Lemma 06N7. Choose a quasi-inverse $b^{-1} : \mathcal{X}'' \to \mathcal{X}$ in the 2-category of categories over $\mathcal{C}$. Since the right triangle of the diagram is 2-commutative we see that

$$\begin{array}{ccc} \mathcal{X} & \xleftarrow{b^{-1}} & \mathcal{X}'' \\ {\scriptstyle F}\downarrow & & \downarrow{\scriptstyle g} \\ \mathcal{Y} & \xleftarrow{} & \mathcal{Y} \end{array}$$

is 2-commutative. Hence we obtain a 1-morphism $c : \mathcal{X}'' \to \mathcal{X} \times_{F,\mathcal{Y},\mathrm{id}} \mathcal{Y}$ by the universal property of the 2-fibre product. Moreover c is a morphism of categories over $\mathcal{Y}$ (!) and an equivalence (by the assumption that b is an equivalence, see Lemma 02XB). Hence c is an equivalence in the 2-category of categories fibred in groupoids over $\mathcal{Y}$ by Lemma 003Z.

We still have to construct a 2-isomorphism between $c \circ b$ and the functor $d : \mathcal{X} \to \mathcal{X} \times_{F,\mathcal{Y},\mathrm{id}} \mathcal{Y}$, $x \mapsto (p(x), x, F(x), \mathrm{id}_{F(x)})$ constructed in the proof of Lemma 06N7. Let $\alpha : F \to g \circ b$ and $\beta : b^{-1} \circ b \to \mathrm{id}$ be 2-isomorphisms between 1-morphisms of categories over $\mathcal{C}$. Note that $c \circ b$ is given by the rule

$$x \mapsto (p(x), b^{-1}(b(x)), g(b(x)), \alpha_x \circ F(\beta_x))$$

on objects. Then we see that

$$(\beta_x, \alpha_x) : (p(x), x, F(x), \mathrm{id}_{F(x)}) \longrightarrow (p(x), b^{-1}(b(x)), g(b(x)), \alpha_x \circ F(\beta_x))$$

is a functorial isomorphism which gives our 2-morphism $d \to b \circ c$. Finally, if the diagram commutes then α_x is the identity for all x and we see that this 2-morphism is a 2-morphism in the 2-category of categories over $\mathcal{Y}$. □